

Discrete Mathematics
MATH1031 - Durham University

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Part I

Module Content

13 Introduction to Discrete Mathematics

13.1 Arrangements and Permutations

13.1.1 Arrangements

Definition 13.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 13.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 13.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

Standard Problem 2: How many arrangements of length k with no repeats can be made from elements of a set S of size $|S| = n$?

Solution:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}, \quad k \leq n \quad (1)$$

As we look at the available choices for each successive item in the list, the number of options is reduced by one each time.

Definition 13.1.1.3: The multiplication principle states that, for an arrangement of length k , if for $i \in \{1, 2, \dots, k\}$ there are n_i choices for the n^{th} item, regardless of previous choices, then the number of possible arrangements is $n_1 n_2 \dots n_k$.

13.1.2 Permutations

Definition 13.1.2.1: A permutation on a finite set S is an arrangement of length $|S|$ that uses each item from S exactly once.

Proposition 13.1.2.1: If $|S| = n$, then there are $n!$ permutations on S .

Proof: This is Standard Problem 2 with $n = k$.

□

Definition 13.1.2.2: We generalise this concept as, for a non-negative integer r , an r -permutation on S is an arrangement of r distinct elements from S . We write $P(n, r)$ for the number of r -permutations on a set of size n , which, by Standard Problem 2, we have,

$$P(n, r) = n(n-1)\dots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n \quad (2)$$

13.1.3 Counting Examples

Definition 13.1.3.1: The addition principle states that, for two disjoint, finite sets A and B . Then,

$$A \cap B = \emptyset \quad |A \cup B| = |A| + |B| \quad (3)$$

More generally, if $A_1, A_2, \dots, A_n$ are pairwise disjoint, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| \quad (4)$$

13.2 Combinations

Definition 13.2.0.1: A combination from a set of objects is an unordered selection of objects of that set.

Standard Problem 3: How many combinations without repetition do we have of a given size k from a set of size n ?

Solution: First consider that there are $P(n, k) = \frac{n!}{(n-k)!}$ arrangements without repetition. These are ordered lists, sorting them into groups of lists using the same letters will give groups of size $k!$. This means that the number of combinations is

$$\frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!} \quad (5)$$

Using the binomial coefficient notation, this means the number of combinations is $\binom{n}{k}$.

Example 13.2.0.1: How many ways are there of selecting ten different letters of the alphabet?

Solution: $\binom{26}{10} = \frac{26!}{10!16!}$.

Proposition 13.2.0.1: $\binom{n}{k} = \binom{n}{n-k}$

Proof: This can be seen from the formula for the binomial.

□

Proposition 13.2.0.2: Pascal's formula, for $1 \leq k \leq (n-1)$, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof: Proof excluded.

□

Theorem 13.2.0.1: The binomial theorem, if n is a positive integer and $a, b \in \mathbb{R}$, then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (6)$$

Proof: The expression $(a+b)(a+b)\dots(a+b)$ when multiplied out, gives various terms of the form $a^k b^{n-k}$. For fixed k , the number of terms $a^k b^{n-k}$ is $\binom{n}{k}$ since one chooses the k brackets in the product form from which to choose the factor a rather than b .

□

13.3 Arrangements and Combinations with Repetitions

14 Combinatorics

14.1 The Three Principles

14.2 Recurrence Relations

14.3 Generating Functions

15 Graph Theory

15.1 What is a Graph?

15.2 Eulerian Graphs

15.3 Planar Graphs

Part II

References

16 Module References

16.1 Module Professors

Introduction to Discrete Mathematics and *Combinatorics* are taught by Professor Mikhail Menshikov.

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Graph Theory is taught by Irene Ayuso-Ventura.

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16.2 Cited Works

n/a